

BACHELOR OF DATA SCIENCE

MAT 201

Linear Algebra

LEARNING MODULE 1

Review of Linear Algebra and Geometry:

The Open University of Kenya

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Reviewed by:

Programme Title	Bachelor of Data Science
Course Title	MAT 201 - Linear Algebra
Learning Module number	Module 1
Learning module title	Review of Linear Algebra and Geometry:
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Reviewed by	

1. COURSE LEARNING GUIDE

1.1. Preparing for Online and Distance Learning

This course offers you the opportunity to enter an online space where you can engage with fellow students at the university. The online forums, exercises, tasks and materials have been designed by the course facilitators in such a way that you can draw directly from their knowledge and share in their scholarly interest in online learning and assessment

We hope you will use your access to this virtual classroom to raise all your questions on the course module, to build a network great learners and to share your own thoughts and experiences to the benefit of your fellow students.

There are a number of steps you can take to ensure you are well prepared to make the most of this learning opportunity. This resource should prompt you to consider the following:

- ☐ How can you best prepare the space where you expect you will spend most of your time engaging in online learning (i.e. in your home or office, or even during your commute to work or whilst traveling)?
- ☐ What can you do to ensure you keep working through the course material, even when you do not have Internet access?
- ☐ How can you familiarise yourself with the course layout, in order to be able to navigate through the steps with ease?
- ☐ What are the communication strategies and best practices you could consider to best engage with your fellow students and the course facilitators?
- ☐ What do you do when you need academic or technical support?

1.2. General Module Guide

The authors of this module believes that imparting knowledge should always be a mutual relationship between the students and the teachers. Students can actively construct knowledge through experiences gained in reading and answering the learning modules.

You should take time to read everything written in the module, actively and religiously solve all the exercises included to finish this course and be able to gain all the competencies and skills expected to demonstrate the learning outcomes as stated in this course. If you finish with full confidence that you understand and can apply all the knowledge you gain in this course, then you are sure to succeed in tackling the next higher subjects in your chosen course.

In this course, you will be taught not only the knowledge you need to demonstrate your learning outcomes, but you will also gain discipline, perseverance, higher thinking skills, and resourcefulness that would help strengthen your foundation to succeed in life. To successfully finish the course, you must follow the guides set forth in using this module.

STUDY ENVIRONMENT.

Before engaging into the module, make sure that you are comfortable enough and diminish distractions. Make sure that you develop a study routine to perform and accomplish all the activities set forth by the module.

MODE OF DELIVERY.

Students are required to get access a digitised copy of the module from the Virtue Learning Environment. As an open and distance learner, you should be able to learn independently and optimise the learning modes and environment available to you as much of the delivery of all the lessons will be done asynchronously. Your success will greatly depend on your self-motivation and discipline to get your work done without your instructor to remind you from time to time.

STUDY SCHEDULE.

Remember that you have other subjects to attend to, hence, you must manage your time so that you will not be left behind in any of them. Create your own learning plan for each course to make sure that you can perform all the activities. Do not procrastinate your work, remember that time wasted can never be retrieved back. Make sure that even if you are at home, you are a student enrolled in a program, where the learning of all its courses greatly depends on your will to learn as the materials you need are all in your hands. So always be aware of your daily tasks.

PRACTICE EXERCISE.

To measure your learning from your reading, you are given practice exercises which you are supposed to answer daily. This will prepare you in performing your main task.

ONLINE GROUP CHAT.

I am going to create an online Discussion Forum and QA Forum on the VLE. This will serve as a room for discussion on the topics and activities found in your module.

RECOMMENDED REFERENCES FOR SUPPLEMENTARY READING.

You are given links that you need to access online. These links are online references such as e-books, online activities, and tutorial videos of the topic. These are other references you can use to further understand the topic and gain additional knowledge and skills.

MAIN TASK.

You are given a task that will require you to apply the knowledge and skills you learned from the module. This can be a group or individual activity which will measure your higher order thinking skills you acquired throughout the topic.

RUBRICS.

This is a tool to guide you in performing your main task. Through this, you will be able to know the specific indicators that should be found in your task. This is to assess your constructive behaviour towards your learning application. You will be rated according to a 4-point scale shown below.

4 – Excellent 3 – Good 2 – Fair 1 – Poor

REFLECTIONS.

Writing your reflection will be a passageway to express your experiential learning in using this module. In your honest recall, write everything that you think happened to you during your learning experience in this module. Use the guide questions below:

First paragraph.

How was your learning experience? Maybe you can consider describing the effect of the parts of the module (sequence, discussions, examples, practice exercises, online activities, and the main task) in your learning experience. You can point out the strength and weakness of each part so that we can consider it for future improvement of this instructional material.

Second paragraph.

What did you learn? Did you successfully acquire the learning outcomes?

Third paragraph.

How can you use what you learned in the future?

2. ASSESSMENT TASKS & EXAMINATION RULES AND GUIDELINES

1. SUBMISSION OF ASSESSMENT TASKS.

Since mathematical skills are best acquired through constant practice, you are encouraged to answer at least two (2) exercises a day. These practice exercises will not be graded but it will help you develop your thinking skills which will prepare you in the performance of your main tasks. You are required to submit your practice exercises every week.

Usually submission deadlines are set on Saturday 11:59PM EAT

2. GROUP CHAT.

You can post questions, solutions, and comment to other's post. Please be reminded that our group chat is exclusive only for discussion on topics related to this course. Everyone is required to actively participate in the discussions. Also, trolling is highly discouraged. Always observe proper online etiquettes when participating in discussions.

RULE OF THUMB: ALWAYS THINK BEFORE YOU CLICK

3. MAIN TASK.

Complete documentation during the preparation of your task is required to ensure that every member of the group participated and contributed in the task. May it be photos or screenshots of your virtual meetings or group works. You are going to submit it as attachment of your main task the same way as the practice exercises.

4. RECOMMENDED REFERENCES FOR SUPPLEMENTARY READING.

You can explore online for other references that is related to the topic to gain additional information about the topic. Do not share or post inappropriate materials online or even privately.

2.1. REQUIREMENTS TO PASS THE COURSE

Compiled Continuous Assignment/Activities and Tasks with Final Performance Examination.

Grading System

Continuous Activities and Tasks	50%
Final Performance Examination.	50%

2.2. INTRODUCTORY MESSAGE

A warm welcome to **MAT 201** - Linear Algebra, **Module on Review of Linear Algebra and Geometry**.

By this module learning resource, we aim to enable you achieve the relevant competencies, depict skill, action and purpose successfully.

This module was designed to provide you with fun and meaningful opportunities for guided and independent learning at your own pace and time. You will be enabled to process the contents of the learning material while being an active learner. Your academic success lies in your own hands. This module has the following parts and corresponding icons:

Table 1:

	AUDIENCE	This is a description of a target audience.
	MODULE DESCRIPTION	This part explains the importance of the topics discussed in the module. It will also give you a sneak peek of what you will learn and what is expected from you at the end of the module.
	EXPECTATION	These are what you will be able to know after completing the lessons in the module
	PRE-ACTIVITY	This activity is to get you set, activate and engage schema to learn the module. Through this activity, you will know the relevance of this activity to the topic that will be discussed in the module.
	PRE - ASSESSMENT	This will measure your prior knowledge and the concepts to be mastered throughout the lesson.
	RECAP	This section will measure what learnings and skills that you understand from the previous lesson.
	LESSON	This section will discuss the topic for this module.

Continued on next page

Table 1: (Continued)

	ACTIVITIES	This is a set of activities you will perform.
 Summary	WRAP UP	This section summarizes the concepts and applications of the lessons.
	DISCUSSIONS	This is the meat of the module. The discussion is aligned to the topic learning outcomes that is expected of you to acquire at the end of the module.
	VALUING	This part will check the integration of values in the learning competency you will take home.
	POST - ASSESSMENT	This will measure how much you have learned from the entire module.
	REFLECTION	This part is for you to share your thoughts by writing your reflection, make sure to check your grammar and spelling to ensure the readability and so that it will not alter the thought being delivered.

Audience Description



This module is offered to all students taking the Bachelor of Data Science. As an open and distance learner, you should be able to learn independently and optimise the learning modes and environment available to you. Before you begin this course, please confirm the course material, the course requirements and how the course is conducted.



Module Description

This module provides a comprehensive review of key concepts in linear algebra and geometry, essential for students and professionals in fields such as mathematics, physics, engineering, computer science, and economics. The module covers foundational topics including vectors, dot products, distances, projections, and matrix algebra. It is designed to reinforce understanding and application of these principles in various contexts.



Expectations

This module seeks to enable learning about:

Module Learning Outcomes

By the end of this module, you should be able to:

1. perform basic vector operations such as addition, subtraction, and scalar multiplication.
2. calculate the dot product of two vectors.
3. apply the dot product to solve problems involving orthogonality and vector projections.
4. perform matrix operations including addition, subtraction, scalar multiplication, and matrix multiplication.

3. Course Home Page

3.1. Course Purpose

To introduce the learner to matrix analysis and its applications

3.2. Course Objective

This course seeks to enable learning about:

1. describe matrix computations in relation to various concepts in linear algebra.
2. determine orthogonal bases for vector spaces.
3. construct decompositions of matrices for efficient computations.
4. apply concepts in linear algebra to solve concrete problems.

3.3. Expected Learning Outcomes (ELO)

By the end of this course, the learner should be able to;

- ELO1. describe matrix computations in relation to various concepts in linear algebra.
- ELO2. determine orthogonal bases for vector spaces.
- ELO3. construct decompositions of matrices for efficient computations.
- ELO4. apply concepts in linear algebra to solve concrete problems.

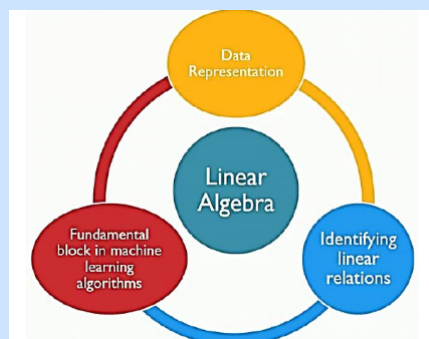
3.4. Course Content

Linear Algebra and Geometry: vectors, dot product, distance, projection, matrix algebra **Solving linear systems of equations:** Geometric approach to the determinant as a measurement of volume; Row operations, echelon form, and geometric interpretation; Degenerate matrices, non-existence of solutions, criterion for existence of solutions; **Vector spaces:** Vector spaces over a field; Basis and dimension; linear independence and spanning sets; **Matrix factorizations:** LU and QR factorizations; Geometric interpretation of linear transformations; Eigenvalues and eigenvectors, diagonalization; Singular Value Decomposition; Principal Component Analysis;

4. INTRODUCTION

Question: Why study Linear Algebra?

The study of linear algebra is indispensable for a prospective student of pure, statistics or applied mathematics. It has great value for any undergraduate student who wishes to go beyond the routine manipulations of formulas to solve standard problems, because it develops the ability to think deductively, analyze mathematical situations, and extend ideas to a new context. The subject of linear algebra is one of the fundamental areas of mathematics, and is the foundation for the study of many advanced topics, not only in mathematics, but also in engineering and the physical sciences. A thorough understanding of the concepts of linear algebra has also become increasingly important for the study of advanced topics in economics and the social sciences. Topics such as systems of linear equations, vector spaces and linear transformations are fundamental in mathematics and physics as well as engineering, economics, and many other areas.



Hello and welcome, you are now at the first part of this module. Your first task is to freshen up on the concepts learned from the previous lessons.



RECAP

In MAT 101: Foundations of Mathematics you learned that a complex number can be represented as a vector. That is, a quantity with both magnitude and direction. Vectors play a very important role, and matrices are an example of functions between sets of vectors. In this module, we will formally introduce the concept of vectors, operations on vectors solve problems involving vectors. It will be important to write these vectors in matrix form to make calculations easier.

In most aspects of life, you will encounter linear equations. In most applications, these equations can be rather large and will need to be converted to matrices so that they can be solved. You will be introduced to matrices, albeit smaller ones, and learn basic numerical and algebraic operations on matrices. Some of the most interesting applications of matrices will be learned at a later stage, but here you will be introduced to basic operations. The most important, and widely applied property of matrices that you will learn in this module is the determinant of a square matrix.

You might have learnt of vectors as quantities with magnitude and direction. In this section, you are going to learn the formal definition of a vector, operations on vectors and study some examples highlighting the properties.



Self Pre-recorded Video 1

A video following the lesson on vectors: their definition and operations on vectors.

Vectors

Definition 4.1. A **vector** is a quantity that has both magnitude and direction. In data analysis, data is frequently arranged in arrays, that is, sets whose elements are indexed. If the data consists of numbers, then a one-dimensional array is called a **vector** i.e., $(a_1, a_2, a_3, \dots, a_n)$ and a two or higher-dimensional array is called a **matrix**. In this section, you will learn about algebraic operations on vectors.

A vector $\mathbf{v}$ in n -dimensional space, $\mathbb{R}^n$, has n components and may be represented as

$$\mathbf{v} = (va_1, a_2, \dots, a_n).$$

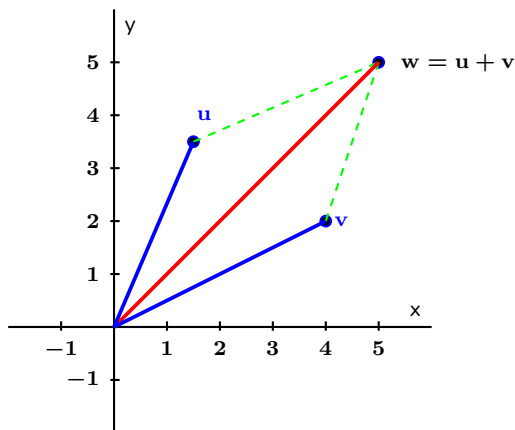
You have already been introduced to complex numbers as vectors $z = (x, y)$ which is in a 2-dimensional space. In this lesson, we will consider the 2-dimensional space and the 3-dimensional space.

Properties of vectors

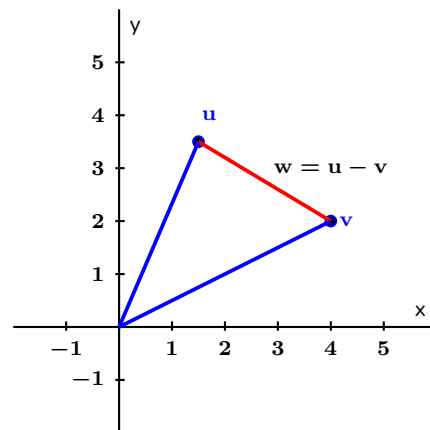
These operations can only be performed on vectors of the same dimension. Let $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ be vectors in $\mathbb{R}^n$ and $a, b \in \mathbb{R}$ be scalars. Then

Commutativity	$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
Associativity	$(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
Additive inverse	$(-\mathbf{v}) + \mathbf{v} = \mathbf{0} = (0, 0, \dots, 0)$
Distributive	$(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$
Distributive	$a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$
Distributive	$(ab)\mathbf{v} = a(b\mathbf{v})$
Multiplicative identity	$1 \cdot \mathbf{v} = \mathbf{v}$

Vectors can be presented geometrically on the cartesian plane as shown below

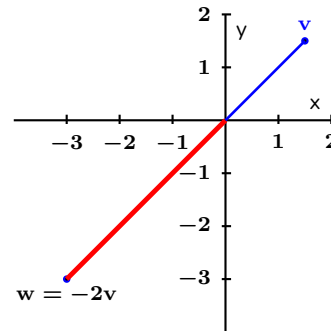
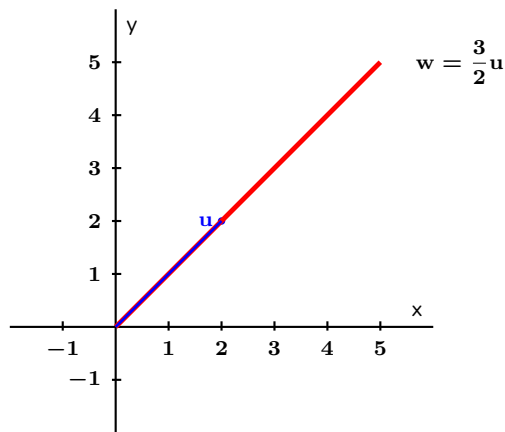


This Shows the sum of two vectors $\mathbf{u}$ and $\mathbf{v}$ to give $\mathbf{w} = \mathbf{u} + \mathbf{v}$



This Shows the difference of two vectors $\mathbf{u}$ and $\mathbf{v}$ to give $\mathbf{w} = \mathbf{u} - \mathbf{v}$

Multiplication of a vector by a non-zero scalar has the effect of either increasing the magnitude, decreasing the magnitude or reversing the direction if the scalar is a negative number. See the graphs below.



This Shows the product of a vector u by a scalar $\frac{3}{2}$ to give $w = \frac{3}{2}u$

This Shows the product of a vector v by a scalar -2 to give $w = -2v$



Example 1:

Let $u = (-2, 3)$, $v = (7, 5)$ and $w = (4, 1)$. Calculate $u + v + w$, $3u - v + 2w$, $4w + 2u$.



Solution

$$u + v + w = (-2, 3) + (7, 5) + (4, 1) = (9, 9)$$

Similarly

$$3u - v + 2w = 3(-2, 3) - (7, 5) + 2(4, 1) = (-5, 6),$$

and

$$4w + 2u = 4(4, 1) + 2(-2, 3) = (12, 10).$$



Example 2:

Let $u = (5, -2, 3)$, $v = (7, 5, 6)$ and $w = (-1, 0, 1)$. Calculate $u + v + w$, $3u - v + 2w$, $4w + 2u$

Solution

$$\mathbf{u} + \mathbf{v} + \mathbf{w} = (5, -2, 3) + (7, 5, 6) + (-1, 0, 1) = (11, 3, 10)$$

Similarly

$$3\mathbf{u} - \mathbf{v} + 2\mathbf{w} = 3(5, -2, 3) - (7, 5, 6) + 2(-1, 0, 1) = (-1, -16, -1),$$

and

$$4\mathbf{w} + 2\mathbf{u} = 4(-1, 0, 1) + 2(5, -2, 3) = (6, -4, 10).$$

Definition 4.2 (Magnitude of a vector). *The magnitude of a vector $\mathbf{v} = (a_1, a_2, \dots, a_n)$ is given by*

$$|\mathbf{v}| = \sqrt{a_1^2 + a_2^2 + \dots + a_n^2}.$$



Example 3:

Let $\mathbf{u} = (-2, 3)$, $\mathbf{v} = (7, 5, 6)$ and $\mathbf{w} = (1, 3, 1)$. Calculate the magnitude of $\mathbf{u}$, $\mathbf{v} + \mathbf{w}$ and $-\mathbf{v} + 2\mathbf{w}$.

Solution

$$|\mathbf{u}| = \sqrt{(-2)^2 + 3^2} = \sqrt{13}$$

$$\mathbf{v} + \mathbf{w} = (7, 5, 6) + (1, 3, 1) = (8, 8, 7)$$

so that

$$|\mathbf{v} + \mathbf{w}| = \sqrt{64 + 64 + 49} = \sqrt{177}.$$

Lastly,

$$-\mathbf{v} + 2\mathbf{w} = -(7, 5, 6) + 2(1, 3, 1) = (-5, 1, -4),$$

so that

$$|-\mathbf{v} + 2\mathbf{w}| = \sqrt{(-5)^2 + 1 + (-4)^2} = \sqrt{42}.$$

1. Let $\mathbf{u} = (-3, 5, 1)$, $\mathbf{v} = (0, 2, 1)$ and $\mathbf{w} = (-1, 2, 1)$. Calculate

(a) $\mathbf{u} - 3\mathbf{v} + 2\mathbf{w}$

(c) $|\mathbf{u} - 3\mathbf{v} + 2\mathbf{w}|$

(b) $4\mathbf{u} - \mathbf{u}$

(d) $4|\mathbf{u}| - |\mathbf{u}|$

2. Let $\mathbf{u} = (-3, 3)$, $\mathbf{v} = (2, -3)$ and $\mathbf{w} = (-1, 2)$. Calculate

- | | |
|--|--|
| (a) $\mathbf{u} - 3\mathbf{v} + 2\mathbf{w}$ | (c) $ \mathbf{u} - 3\mathbf{v} + 2\mathbf{w} $ |
| (b) $4\mathbf{u} - \mathbf{u}$ | (d) $4 \mathbf{u} - \mathbf{u} $ |

Definition 4.3 (Column Vectors). When vectors are written vertically instead of horizontally, then they are called column vectors. An example is

$$\mathbf{v} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}.$$

Operations of addition and scalar multiplication can be performed on column vectors just as was the case with row vectors.

4.4. Inner Products

Up to now we have considered a vector space solely as an entity with an algebraic structure. We know, however, that $\mathbb{R}^n$ possesses more than just an algebraic structure; in particular, we know that we can measure the "size" or "length" of a vector $\bar{\mathbf{x}}$ in $\mathbb{R}^n$ by the quantity $\|\bar{\mathbf{x}}\| = \sqrt{\bar{\mathbf{x}}^T \bar{\mathbf{x}}}$. Similarly, we can define distance from $\bar{\mathbf{x}}$ to $\bar{\mathbf{y}}$ as $\|\bar{\mathbf{x}} - \bar{\mathbf{y}}\|$. The ability to measure distances means that $\mathbb{R}^n$ has a geometric structure, which supplements the algebraic structure. The geometric structure can be employed to study problems of convergence, continuity, and the like.

The geometrical structure of $\mathbb{R}^n$ is based on the scalar product $\bar{\mathbf{x}} \bullet \bar{\mathbf{y}}$ or $\bar{\mathbf{x}}^T \bar{\mathbf{y}}$. Essentially the scalar product is a real-valued function of two vector variables: Given $\bar{\mathbf{x}}$ and $\bar{\mathbf{y}}$ in $\mathbb{R}^n$, the scalar product or dot product produces a number $\bar{\mathbf{x}}^T \bar{\mathbf{y}}$. Thus to derive a geometric structure for a vector space V , we should look for a generalization of the scalar product function.

Definition 7.1 An inner product on a real vector space V is a function that assigns a real number, $\langle \bar{\mathbf{u}}, \bar{\mathbf{v}} \rangle$, to each pair of vectors $\bar{\mathbf{u}}$ and $\bar{\mathbf{v}}$ in V , and that satisfies the following properties.

For any vectors $\bar{\mathbf{u}}, \bar{\mathbf{v}}, \bar{\mathbf{w}}$ and any scalar α ,

1. $\langle \bar{\mathbf{u}}, \bar{\mathbf{u}} \rangle \geq 0$ and $\langle \bar{\mathbf{u}}, \bar{\mathbf{u}} \rangle = 0$ if and only if $\bar{\mathbf{u}} = \bar{\mathbf{0}}$. (Positivity axiom; Definiteness axiom)
2. $\langle \bar{\mathbf{u}}, \bar{\mathbf{v}} \rangle = \langle \bar{\mathbf{v}}, \bar{\mathbf{u}} \rangle$.
(Symmetry axiom)
3. $\langle \alpha \bar{\mathbf{u}}, \bar{\mathbf{v}} \rangle = \alpha \langle \bar{\mathbf{u}}, \bar{\mathbf{v}} \rangle$
(Homogeneity axiom)
4. $\langle \bar{\mathbf{u}}, \bar{\mathbf{v}} + \bar{\mathbf{w}} \rangle = \langle \bar{\mathbf{u}}, \bar{\mathbf{v}} \rangle + \langle \bar{\mathbf{u}}, \bar{\mathbf{w}} \rangle$
(Additivity axiom).

Definition 7.2 Any real vector space with an inner product is called an inner product space (IPS).

The axioms of an inner product imply the following additional basic properties.

Theorem 7.1 Let $\bar{\mathbf{u}}, \bar{\mathbf{v}}$ and $\bar{\mathbf{w}}$ be any vectors in an inner product space and let α be a scalar. Then

1. $\langle \bar{u}, \bar{v} + \bar{w} \rangle = \langle \bar{u}, \bar{v} \rangle + \langle \bar{u}, \bar{w} \rangle$
2. $\langle \bar{u}, \alpha \bar{v} \rangle = \alpha \langle \bar{u}, \bar{v} \rangle$
3. $\langle \bar{u} - \bar{w}, \bar{v} \rangle = \langle \bar{u}, \bar{v} \rangle - \langle \bar{w}, \bar{v} \rangle$
4. $\langle \bar{u}, \bar{v} - \bar{w} \rangle = \langle \bar{u}, \bar{v} \rangle - \langle \bar{u}, \bar{w} \rangle$
5. $\langle \bar{0}, \bar{v} \rangle = \langle \bar{v}, \bar{0} \rangle = 0$

Examples.

1. The usual scalar or dot product in $\mathbb{R}^n$ is an inner product where $\langle \bar{x}, \bar{y} \rangle = \bar{x}^T \bar{y} = x_1 y_1 + x_2 y_2 + \dots + x_n y_n$, where $\bar{x} = (x_1, x_2, \dots, x_n)$ and $\bar{y} = (y_1, y_2, \dots, y_n)$.
2. Let V be the vector space $\mathbb{R}^2$ and let A be the 2×2 matrix $A = \begin{pmatrix} 3 & 2 \\ 2 & 4 \end{pmatrix}$. Verify that the function $\langle \bar{u}, \bar{v} \rangle = \bar{u}^T A \bar{v}$ is an inner product for $\mathbb{R}^2$.
3. Let $\mathbb{P}_2$ denote the vector space of polynomials of degree 2 or less. Let $p(t), q(t) \in \mathbb{P}_2$. Verify that

$$\langle p, q \rangle = \int_0^1 p(t)q(t)dt$$

is an inner product.

4. Let $u = (u_1, u_2)$ and $v = (v_1, v_2)$ be any 2 -vectors. Then

$$\langle u, v \rangle = 3u_1v_1 + 4u_2v_2$$

defines an inner product.

note that a vector space may have several different inner products.

Definition 7.3 The norm(or length or magnitude) of a vector $\bar{v}$ is the nonnegative number $\|\bar{v}\|$ defined by $\|\bar{v}\| = \sqrt{\langle \bar{v}, \bar{v} \rangle}$.

Remark. A norm on a vector space is a real-valued function. In a real vector space, define the angle θ between two nonzero vectors $\bar{u}, \bar{v}$ by

$$\cos \theta = \frac{\langle \bar{u}, \bar{v} \rangle}{\|\bar{u}\| \|\bar{v}\|}, \quad 0 \leq \theta \leq \pi$$

Definition 7.4 Let V be an inner product space. Two vectors $\bar{u}$ and $\bar{v}$ are orthogonal if their inner product is zero. That is, if $\langle \bar{u}, \bar{v} \rangle = 0$.

Note that the concept of angle and orthogonality depends on the inner product being used. In a real vector space with nonzero $\bar{u}$ and $\bar{v}$, $\bar{u} \perp \bar{v} \iff \theta = 90^\circ$.

Matrices play a very important role in data analysis. Huge data sets are stored in the form of matrices for easy analysis. In this lesson, you will be introduced to matrices, their definition, operation of addition, linear combinations and multiplication of matrices. Pay special attention to multiplication of matrices!



Self Pre-recorded Video 1

A video following the lesson on matrices, the dimension of matrices linear operations on matrices and their properties and multiplication of matrices.

Matrices

Definition 4.5. A matrix A is an $m \times n$ rectangular array of numbers usually presented in the form

$$A_{m \times n} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Expressed in this form, the matrix has m rows and n columns. We call this matrix an $m \times n$ matrix. A matrix with only one row is called a **row matrix** or **row vector**, and a matrix with only one column is called a **column matrix** or **column vector**. A matrix whose entries are all zero is called a **zero matrix** and will usually be denoted by 0 .



Example 4:

Here are some examples of matrices.

$$A = \begin{bmatrix} 1 & 3 & -1 \\ 4 & -2 & 5 \end{bmatrix}, B = \begin{bmatrix} 1 & 3 \\ -1 & 4 \\ -2 & 5 \end{bmatrix}, C = \begin{bmatrix} 1 & 3 & -1 \\ 4 & -2 & 5 \\ 1 & 7 & 6 \end{bmatrix}, D = \begin{bmatrix} 3 & -1 \\ 4 & -2 \end{bmatrix}$$

A is a 2×3 matrix, B a 3×2 matrix, C a 3×3 matrix and D a 2×2 matrix. C and D are examples of square matrices.

Matrix operations

Matrices of the same dimension behave exactly like vectors when performing operations of addition, subtraction and scalar multiplication. Properties of vectors discussed in the previous section apply as well.



Example 5:

Given that $A = \begin{bmatrix} 1 & 3 & -1 \\ 4 & -2 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 2 & -1 & 4 \\ -1 & 4 & -2 \end{bmatrix}$, evaluate $A+B$ and $2A-3B$.



Solution

$$\begin{aligned} A + B &= \begin{bmatrix} 1 & 3 & -1 \\ 4 & -2 & 5 \end{bmatrix} + \begin{bmatrix} 2 & -1 & 4 \\ -1 & 4 & -2 \end{bmatrix} \\ &= \begin{bmatrix} 1+2 & 3+(-1) & -1+4 \\ 4+(-1) & -2+4 & 5+(-2) \end{bmatrix} \\ &= \begin{bmatrix} 3 & 2 & 3 \\ 3 & 2 & 3 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} 2A - 3B &= 2 \begin{bmatrix} 1 & 3 & -1 \\ 4 & -2 & 5 \end{bmatrix} - 3 \begin{bmatrix} 2 & -1 & 4 \\ -1 & 4 & -2 \end{bmatrix} \\ &= \begin{bmatrix} 2+(-6) & 6+(3) & -2+(-12) \\ 8+(3) & -4+(-12) & 10+(6) \end{bmatrix} \\ &= \begin{bmatrix} -4 & 9 & -14 \\ 11 & -16 & 16 \end{bmatrix} \end{aligned}$$



Example 6:

Given that $M = \begin{bmatrix} 3 & -1 \\ 2 & -1 \\ 4 & -2 \end{bmatrix}$, $N = \begin{bmatrix} 5 & -1 \\ 2 & 4 \\ 4 & -2 \end{bmatrix}$, evaluate $2N - M$ and $-8N$.

Solution

$$\begin{aligned}
 2N - M &= 2 \begin{bmatrix} 5 & -1 \\ 2 & 4 \\ 4 & -2 \end{bmatrix} - \begin{bmatrix} 3 & -1 \\ 2 & -1 \\ 4 & -2 \end{bmatrix} \\
 &= \begin{bmatrix} 10 - 3 & -2 - 1 \\ 4 - 2 & 8 - (-1) \\ 8 - 4 & -4 - (-2) \end{bmatrix} \\
 &= \begin{bmatrix} 7 & -3 \\ 2 & 9 \\ 4 & 2 \end{bmatrix}
 \end{aligned}$$

and

$$\begin{aligned}
 -8N &= -8 \begin{bmatrix} 5 & -1 \\ 2 & 4 \\ 4 & -2 \end{bmatrix} \\
 &= \begin{bmatrix} -40 & 8 \\ -16 & -32 \\ -32 & 16 \end{bmatrix}
 \end{aligned}$$

1. Let $A = \begin{bmatrix} 3 & 1 & -1 \\ 2 & 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 3 & 0 \\ 4 & 4 & 1 \end{bmatrix}$, evaluate $2A + 3B$ and $4A - B$.
2. Given that $M = \begin{bmatrix} 1 & 0 & 5 \\ -1 & 3 & 2 \\ 3 & 1 & -1 \end{bmatrix}$, $N = \begin{bmatrix} 2 & 2 & -1 \\ 4 & -1 & 2 \\ 0 & 4 & 0 \end{bmatrix}$, evaluate $3N + N$ and $5M - 2N$.

Matrix Multiplication

Definition 4.6. Given a row matrix $A = \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix}$ and a column matrix $B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$,

the product AB is defined as

$$AB = \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = a_1b_1 + a_2b_2 + \cdots + a_nb_n.$$

This is not a matrix but a number.

In general, to multiply two matrices A and B, the number of columns of matrix A must be equal to the number of rows of matrix B for the multiplication to make sense. To get the product AB, rows of A are multiplied with the columns of B to get the entries of the matrix AB. See the example below.



Example 7:

Let $A = \begin{bmatrix} 1 & 0 \\ 3 & 2 \\ 1 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 1 & -1 \\ 2 & 1 & 1 \end{bmatrix}$. Evaluate the products AB and BA.

Solution

The dimensions of B are 3×2 and the dimensions of A are 2×3 . The inner dimensions match so the product is defined and will be a 3×3 matrix.

$$\begin{aligned} AB &= \begin{bmatrix} 1 & 0 \\ 3 & 2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3 & 1 & -1 \\ 2 & 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 3 & 1 & -1 \\ 13 & 5 & -1 \\ 1 & 0 & -2 \end{bmatrix}. \end{aligned}$$

On the other hand, the dimensions of A are 2×3 and the dimensions of B are 3×2 . The inner dimensions match so the product is defined and will be a 2×2 matrix.

$$\begin{aligned} BA &= \begin{bmatrix} 3 & 1 & -1 \\ 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 3 & 2 \\ 1 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 5 & 3 \\ 6 & 1 \end{bmatrix} \end{aligned}$$

Notice that AB and BA are not equal! This illustrates the fact that matrix multiplication is NOT commutative.

Definition 4.7 (Properties of Matrix Multiplication). *For matrices A, B and C, the following properties hold.*

1. *Matrix multiplication is Associative:* $(AB)C = A(BC)$.
2. *Matrix multiplication is distributive*

$$\begin{aligned} A(B + C) &= AB + AC \\ (A + B)C &= AC + BC \end{aligned}$$

As we have observed from the previous example. matrix multiplication is NOT commutative.

1. Given the matrices $A = \begin{bmatrix} 2 & 3 \\ 3 & -2 \\ -4 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 6 \\ -3 & 7 \\ 6 & 5 \end{bmatrix}$ and $C = \begin{bmatrix} -1 & -1 & 3 \\ -3 & -2 & 2 \end{bmatrix}$ Evaluate
- AC and CA ,
 - $(A + B)C$ and $C(A + B)$
 - BC and CB
 - Verify that $(A + B)C = AC + BC$ and $C(A + B) = CA + CB$
2. Given the square matrices A , B and C below, evaluate AB , BA , $B(A + C)$ and $A(B + C)$ and $(A + B)C$
- $A = \begin{bmatrix} 6 & 6 \\ 3 & -2 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 7 \\ 10 & -2 \end{bmatrix}$ and $C = \begin{bmatrix} 4 & 2 \\ 5 & 1 \end{bmatrix}$.
 - $A = \begin{bmatrix} 8 & 5 & 2 \\ 2 & 7 & 2 \\ 5 & 6 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 10 & 6 \\ 5 & -1 & 7 \\ 2 & 2 & 10 \end{bmatrix}$ and $C = \begin{bmatrix} 0 & 6 & 10 \\ -3 & 0 & 10 \\ -1 & 0 & -5 \end{bmatrix}$.

Square matrices are very special. Each square matrix can be evaluated for the determinant, which tells us if the matrix has an inverse or not. In this section, you will learn about various ways of calculating the determinant of a square matrix, and properties of determinants.



Self Pre-recorded Video 1

A video following the lesson on calculation of determinants. Use examples to explain calculation of the determinant using row operations, and using the cofactor method.

The determinant of a square matrix

Definition 4.8. Let A be a square $n \times n$ matrix. The determinant of A denoted by $|A|$ or $\det(A)$ is a real number and contains important information about the matrix. We can think of the **determinant as a function**

$$\det : \{\text{Set of square matrices}\} \rightarrow \mathbb{R}$$

The determinant satisfies the following properties

- Doing a row replacement on A (that is a linear operation on the rows of A) does not change $\det(A)$.
- Scaling a row of A (i.e. multiplying a row by a non-zero constant) by a scalar c multiplies the determinant by c .
- Swapping two rows of a matrix multiplies the determinant by -1 .
- The determinant of the identity matrix I_n is equal to 1.
- If the determinant of A is non-zero, then A has an inverse A^{-1} and $\det(A^{-1}) = \frac{1}{\det(A)}$.
- If A and B are two matrices of the same dimension, then $\det(AB) = \det(A)\det(B)$.

There are several ways of evaluating the determinant of square matrices. It is important to know how this is done by hand, but in real life situations, the matrices can be very large, and so calculations are done using Computer Algebra Engines like python, sage, mathematica, matlab, maxima e.t.c.

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be a non zero matrix. Then

$$\det(A) = ab - cd.$$

For ways of evaluating determinants of higher dimensional matrices, see the reading activity below.

Activity 1: Reading activity

Read [?, Section 4] on definition of the determinant and the calculations of determinants using row operations and using the cofactor method. Then attempt the problems below.



Example 8:

Let $A = \begin{bmatrix} 5 & 6 \\ 3 & -2 \end{bmatrix}$. Evaluate the determinant of A.



Solution

$$\det(A) = 5(-2) - 3(6) = -28.$$



Example 9:

Let $A = \begin{bmatrix} 3 & -2 \\ 5 & 6 \end{bmatrix}$. Evaluate the determinant of A.



Solution

$$\det(A) = 3(6) - 5(-2) = 28.$$

From the two examples above, we can see that interchanging the rows of A has the effect of changing the sign of the determinant.

**Example 10:**

Use the cofactor method to evaluate the determinant $\det(C)$ of $C = \begin{bmatrix} 2 & 6 & 5 \\ 3 & 7 & -1 \\ 4 & 0 & 9 \end{bmatrix}$

**Solution**

Using the cofactor method,

$$\begin{aligned} \begin{vmatrix} 2 & 6 & 5 \\ 3 & 7 & -1 \\ 4 & 0 & 9 \end{vmatrix} &= 2 \begin{vmatrix} 7 & -1 \\ 0 & 9 \end{vmatrix} - 6 \begin{vmatrix} 3 & -1 \\ 4 & 9 \end{vmatrix} + 5 \begin{vmatrix} 3 & 7 \\ 4 & 0 \end{vmatrix} \\ &= 2(63 - 0) - 6(27 - (-4)) + 5(0 - 28) \\ &= -200 \end{aligned}$$

**Example 11:**

Use the cofactor method to evaluate the determinant $\det(C)$ of $C = \begin{bmatrix} 2 & 6 & 5 \\ 3 & 7 & -1 \\ 4 & 0 & 9 \end{bmatrix}$

**Solution**

Using the cofactor method,

$$\begin{aligned} \begin{vmatrix} 2 & -2 & 4 \\ 6 & 1 & 7 \\ -1 & 4 & -1 \end{vmatrix} &= 2 \begin{vmatrix} 1 & 7 \\ 4 & -1 \end{vmatrix} - (-2) \begin{vmatrix} 6 & 7 \\ -1 & -1 \end{vmatrix} + 4 \begin{vmatrix} 6 & 1 \\ -1 & 4 \end{vmatrix} \\ &= 2(-1 - 28) + 2(-6 + 7) + 4(24 + 1) \\ &= 44 \end{aligned}$$

Evaluate the determinants of the following matrices

1. $\begin{bmatrix} 5 & 2 \\ -3 & 1 \end{bmatrix},$

2. $\begin{bmatrix} 10 & 0 & -3 \\ 2 & -3 & 7 \\ 2 & 1 & -5 \end{bmatrix},$

3. $\begin{bmatrix} -4 & 4 & 0 \\ 6 & 3 & 2 \\ -5 & -3 & -1 \end{bmatrix}$



If we think of matrices as functions (mappings) between sets, then matrices with non-zero determinants play a key role because they have inverses. Determinants of these matrices can then be used to calculate the area and volume of geometric objects. You will find this application very interesting indeed!



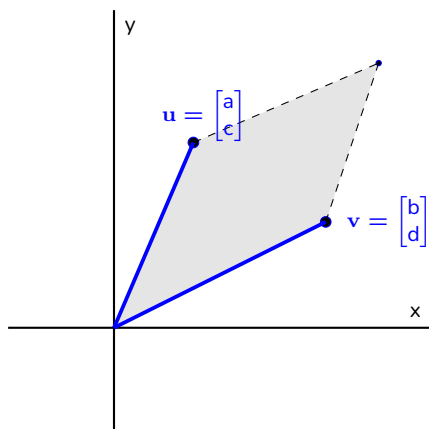
Self Pre-recorded Video 1

A video following the application of determinants in calculating the area of geometric shapes. Of consideration for our context is the area of a parallelogram and the volume of parallelepiped.

Applications of the determinant

Determinants of matrices has very many applications. One of the applications is in solving systems of linear equations, which you will encounter in a later course. We are going to look at the application in evaluating the area and volume of geometric shapes.

Consider the parallelogram in $\mathbb{R}^2$ whose sides are vectors $\mathbf{u} = \begin{bmatrix} a \\ c \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} b \\ d \end{bmatrix}$ in the diagram below



The area of the parallelogram is given as the absolute value of the determinant of the matrix

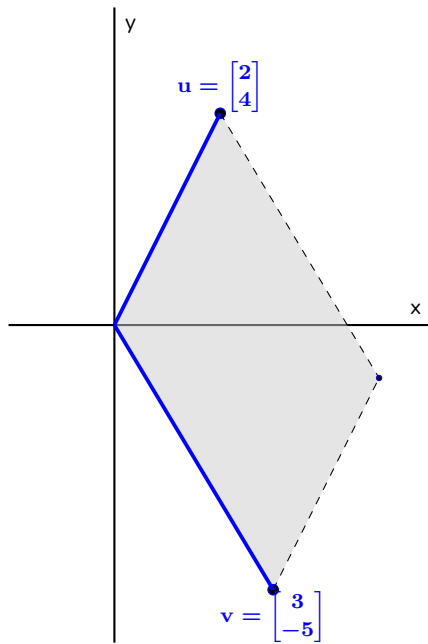
$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. That is

$$\text{Area} = |\det(A)| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = |ad - cb|.$$



Example 12:

Calculate the area of the parallelogram with sides $(2, 4)$ and $(3, -5)$. See the diagram below.

 **Solution**

Here, form the matrix $A = \begin{bmatrix} 2 & 3 \\ 4 & -5 \end{bmatrix}$. The area of the parallelogram will be

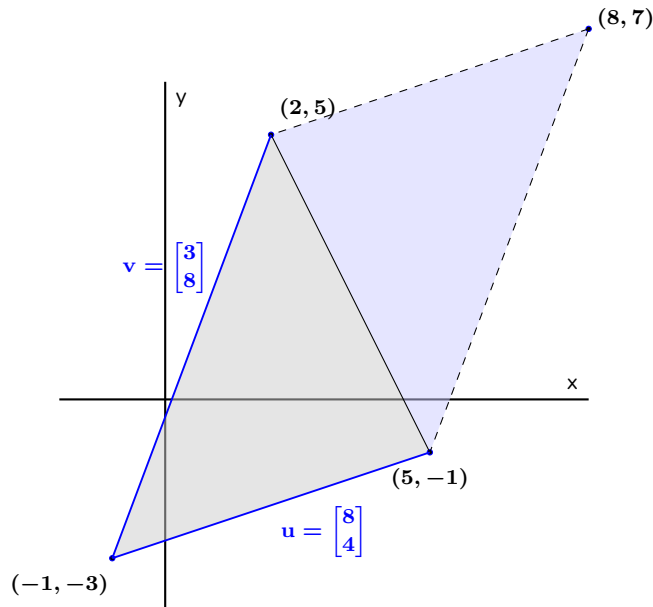
$$\text{Area} = |\det(A)| = \begin{vmatrix} 2 & 3 \\ 4 & -5 \end{vmatrix} = |-10 - 12| = 22 \text{ Square units.}$$

**Example 13:**

Calculate the area of the triangle with vertices at $(-1, -3)$, $(2, 5)$ and $(5, -1)$.

Solution

This is just one half of a parallelogram with vertices $(-1, -3)$, $(2, 5)$, $(5, -1)$ and $(8, 7)$. See the diagram below.



We express the sides of the parallelogram as vectors $\mathbf{u} = \begin{bmatrix} 8 \\ 4 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 3 \\ 8 \end{bmatrix}$. The area of the triangle will be given by

$$\text{Area} = \frac{1}{2} |\det(\mathbf{A})| = \frac{1}{2} \begin{vmatrix} 3 & 8 \\ 8 & -1 \end{vmatrix} = \frac{1}{2} |-3 - 40| = 21.5 \text{ Square units.}$$

More applications of determinants in the calculations of areas and volumes can be found in the reading activity below.

Activity 2: Reading activity

Read on [?, Section 4] on the applications of determinants in calculating areas and volumes then attempt the exercise below.

1. Calculate the area of the triangle with vertices at

(a) $(-1, -2)$, $(3, 8)$ and $(9, 0)$.

(b) $(-5, 5)$, $(5, 4)$ and $(3, -3)$.

2. Calculate the area of a parallelogram whose vertices are given by $(-5, 5)$, $(-4, -3)$, $(3, -2)$ and $(2, 6)$.



RECOMMENDED SUPPLEMENTARY READING

You are required to watch the videos on the basic concepts of vectors, matrix algebra and determinants. Please click on the link below:



Content Curated Video



- on vector operations <https://youtu.be/DxwT7fn6GtE?si=zB-nBakrRmaRBPBw>
- on matrix operations <https://youtu.be/p48uw2vFWQs?si=vFK7PqHCbLwebkL5>
- on determinants <https://youtu.be/3ROzG6n4yMc?si=Vm8DdmvnbEnTkcYR>
- on application of determinants <https://youtu.be/RcAoqXEg3G4?si=2clPd6mfzfcXGBu>



CORE READING AND REFERENCES

1. Vectors. (2020, December 21). Virginia Military Institute. <https://math.libretexts.org/@go/page/4219>
2. Denton, D. C., Tom, Waldron, A. (2021, March 5). Determinants. University of California, Davis.
3. <https://math.libretexts.org/@go/page/1730>
4. Determinants. (2021, March 5). Brigham Young University. <https://math.libretexts.org/@go/page/14516>

Solve the following problems to complete your assessment for this module.

1. Let $\mathbf{u} = (-3, 5, 1)$, $\mathbf{v} = (0, 2, 1)$ and $\mathbf{w} = (-1, 2, 1)$. Calculate $7\mathbf{u} - 3\mathbf{v} - \mathbf{w}$ and $|\mathbf{u} - 3\mathbf{v} + 2\mathbf{w}|$
2. Let $C = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$ and $D = \begin{bmatrix} 5 & 3 \\ 4 & 4 \end{bmatrix}$, evaluate $2A + 3B$ and $4A - B$.
3. Given the matrices $A = \begin{bmatrix} 2 & 3 \\ 3 & -2 \\ -4 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 6 \\ -3 & 7 \\ 6 & 5 \end{bmatrix}$ and $C = \begin{bmatrix} -1 & -1 & 3 \\ -3 & -2 & 2 \end{bmatrix}$ Evaluate AC and CA , $(A + B)C$ and $C(A + B)$.
4. Evaluate the determinants of the following matrices
 - (a) $\begin{bmatrix} 5 & 2 \\ -3 & 1 \end{bmatrix}$,
 - (b) $\begin{bmatrix} 10 & 0 & -3 \\ 2 & -3 & 7 \\ 2 & 1 & -5 \end{bmatrix}$,
5. Calculate the area of the triangle with vertices at $(3, -3)$, $(-4, -3)$ and $(-1, 2)$.



WRAP UP

Summary

To wrap-up, answer the following questions:

1. What is a vector?
2. What is a matrix?
3. Can two matrices be multiplied? What properties do they need to satisfy for the multiplication to make sense?
4. Is the product of two matrices commutative?
5. What is the determinant of a square matrix, and how is it calculated?
6. How can the determinant be used to calculate the area of geometric objects?



VALUING AND TAKE HOME



Hello again, you are now at the last part of this module. Your task is to do a self-reflection on your learning experience.

Vectors are important mathematical objects and so are matrices. You will be encountering matrices and vectors in other courses and in your day to day life. Knowing the definitions and properties helps in the understanding of these mathematical objects. In real life, you will be dealing with very large vectors and matrices of very high dimensions. Some of the calculations done in this course will not be possible by hand, and so you will need to use computer algebra programs to do your calculations. It will be important to have an intuition of what these operations will entail.



DISCUSSION AND REFLECTION



Welcome to the online discussion forum for our module! This is a space for you to ask questions, share insights, and collaborate with your fellow learners as we conquer the world of linear equations. Here are some prompts to get us started:

1. What are your specific areas of interest within linear algebra or geometry?
2. Have you encountered any challenges in the past with these concepts?

3. Are there any real-world examples you'd like to explore where these concepts come into play?

Feel free to share your thoughts, ask questions, and help each other solidify these fundamental mathematical concepts!



CORE READING AND REFERENCES

References

- [1] D. Lay, S. Lay and J. McDonald, 2021, Linear Algebra and Its Applications, Global Edition 6th Edition, Pearson, ISBN-13: 978-1292351216
- [2] D. Lay, S. Lay and J. McDonald, 2020, Study guide for Linear Algebra and Its Applications, 6th Edition, Pearson, ISBN-13: 978-0135851234
- [3] Mike Cohen, 2022, Practical Linear Algebra for Data Science: From Core Concepts to Applications Using Python, O'Reilly Media, ISBN-13: 978-1098120610

To complete this module, take time end of module assessment below. This will contribute to your overall assessment. The test is worth 25 marks



VALUING AND TAKE HOME

Hello again, you are now at the last part of this module. Your task is to do a self-reflection on your learning experience. Here are some prompts to guide your reflection:



- Briefly summarize the 2-3 most important concepts you learned in this module.
- Can you think of any real-world applications or connections to other mathematical concepts you've encountered?
- Did you encounter any specific challenges during this module? If so, how did you overcome them?
- Briefly describe a practice problem you found particularly insightful or challenging. How did solving it deepen your understanding?

- What are you most interested in learning in the next module that builds upon this topic?
- Briefly explain a concept from this module to someone who hasn't taken the course.

Review of Linear Algebra and Geometry:

5 minute review

Recap the definition of a vector, vector operations and their properties, Definition of matrices and operations on matrices. Determinants and their applications to calculating the area of geometric objects.

Problems.

Attempt the problems below for your practice and suggest problems you would want to be discussed in the weekly tutorial session.

1. Vectors and operations on vectors

1. Let $\mathbf{u} = (-3, 5, 1)$, $\mathbf{v} = (0, 2, 1)$ and $\mathbf{w} = (-1, 2, 1)$. Calculate

- | | | |
|--|--|--|
| (a) $\mathbf{u} + \mathbf{v} + \mathbf{w}$ | (c) $4\mathbf{u} - \mathbf{u}$ | (e) $ \mathbf{u} - 3\mathbf{v} + 2\mathbf{w} $ |
| (b) $\mathbf{u} - 3\mathbf{v} + 2\mathbf{w}$ | (d) $ \mathbf{u} + \mathbf{v} + \mathbf{w} $ | (f) $4 \mathbf{u} - \mathbf{u} $ |

2. Let $\mathbf{u} = (-3, 3)$, $\mathbf{v} = (2, -3)$ and $\mathbf{w} = (-1, 2)$. Calculate

- | | | |
|--|--|--|
| (a) $\mathbf{u} + \mathbf{v} + \mathbf{w}$ | (c) $4\mathbf{u} - \mathbf{u}$ | (e) $ \mathbf{u} - 3\mathbf{v} + 2\mathbf{w} $ |
| (b) $\mathbf{u} - 3\mathbf{v} + 2\mathbf{w}$ | (d) $ \mathbf{u} + \mathbf{v} + \mathbf{w} $ | (f) $4 \mathbf{u} - \mathbf{u} $ |

2. Matrix operations

1. Given that $M = \begin{bmatrix} 1 & 0 \\ 3 & 2 \\ 1 & -1 \end{bmatrix}$, $N = \begin{bmatrix} 2 & -1 \\ -1 & 2 \\ 0 & 4 \end{bmatrix}$, evaluate $3N + N$ and $5M - 2N$.

2. Let $A = \begin{bmatrix} 3 & 1 & -1 \\ 2 & 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 3 & 0 \\ 4 & 4 & 1 \end{bmatrix}$, evaluate $2A + 3B$ and $4A - B$.

3. Let $C = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$ and $D = \begin{bmatrix} 5 & 3 \\ 4 & 4 \end{bmatrix}$, evaluate $2A + 3B$ and $4A - B$.

4. Given that $M = \begin{bmatrix} 1 & 0 & 5 \\ -1 & 3 & 2 \\ 3 & 1 & -1 \end{bmatrix}$, $N = \begin{bmatrix} 2 & 2 & -1 \\ 4 & -1 & 2 \\ 0 & 4 & 0 \end{bmatrix}$, evaluate $3N + N$ and $5M - 2N$.

5. Given the matrices $A = \begin{bmatrix} 2 & 3 \\ 3 & -2 \\ -4 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 6 \\ -3 & 7 \\ 6 & 5 \end{bmatrix}$ and $C = \begin{bmatrix} -1 & -1 & 3 \\ -3 & -2 & 2 \end{bmatrix}$ Evaluate

- (a) AC and CA ,
(b) $(A + B)C$ and $C(A + B)$

- (c) BC and CB
 (d) Verify that $(A + B)C = AC + BC$ and $C(A + B) = CA + CB$
6. Given the matrices $A = \begin{bmatrix} 7 & -3 & 3 \\ 7 & 5 & -3 \end{bmatrix}$, $B = \begin{bmatrix} -2 & 9 \\ -3 & 3 \\ 6 & 5 \end{bmatrix}$ and $C = \begin{bmatrix} 3 & -2 & -3 \\ -4 & -3 & 0 \end{bmatrix}$
- Evaluate
- (a) AB and BA ,
 (b) $(A + C)B$ and $B(A + C)$
 (c) BC and CB
 (d) Verify that $(A + C)B = AB + CB$ and $B(A + C) = BA + BC$
7. Given the square matrices A , B and C below, evaluate AB , BA , $B(A + C)$ and $A(B + C)$ and $(A + B)C$
- (a) $A = \begin{bmatrix} 6 & 6 \\ 3 & -2 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 7 \\ 10 & -2 \end{bmatrix}$ and $C = \begin{bmatrix} 4 & 2 \\ 5 & 1 \end{bmatrix}$.
- (b) $A = \begin{bmatrix} 1 & -1 \\ 2 & -3 \end{bmatrix}$, $B = \begin{bmatrix} -2 & -4 \\ 5 & -2 \end{bmatrix}$ and $C = \begin{bmatrix} 5 & 2 \\ -3 & 1 \end{bmatrix}$.
- (c) $A = \begin{bmatrix} 10 & 0 & -3 \\ 2 & -3 & 7 \\ 2 & 1 & -5 \end{bmatrix}$, $B = \begin{bmatrix} -4 & 4 & 0 \\ 6 & 3 & 2 \\ -5 & -3 & -1 \end{bmatrix}$ and $C = \begin{bmatrix} -2 & 2 & -3 \\ 0 & 6 & 1 \\ 3 & -4 & 1 \end{bmatrix}$.
- (d) $A = \begin{bmatrix} 8 & 5 & 2 \\ 2 & 7 & 2 \\ 5 & 6 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 10 & 6 \\ 5 & -1 & 7 \\ 2 & 2 & 10 \end{bmatrix}$ and $C = \begin{bmatrix} 0 & 6 & 10 \\ -3 & 0 & 10 \\ -1 & 0 & -5 \end{bmatrix}$.

3. Calculation of determinants

Evaluate the determinants of the following matrices

1. $\begin{bmatrix} 6 & 6 \\ 3 & -2 \end{bmatrix}$,
2. $\begin{bmatrix} -1 & 7 \\ 10 & -2 \end{bmatrix}$,
3. $\begin{bmatrix} 4 & 2 \\ 5 & 1 \end{bmatrix}$,
4. $\begin{bmatrix} 1 & -1 \\ 2 & -3 \end{bmatrix}$,
5. $\begin{bmatrix} -2 & -4 \\ 5 & -2 \end{bmatrix}$,
6. $\begin{bmatrix} 5 & 2 \\ -3 & 1 \end{bmatrix}$,
7. $\begin{bmatrix} 10 & 0 & -3 \\ 2 & -3 & 7 \\ 2 & 1 & -5 \end{bmatrix}$,
8. $\begin{bmatrix} -4 & 4 & 0 \\ 6 & 3 & 2 \\ -5 & -3 & -1 \end{bmatrix}$,
9. $\begin{bmatrix} -2 & 2 & -3 \\ 0 & 6 & 1 \\ 3 & -4 & 1 \end{bmatrix}$,
10. $\begin{bmatrix} 8 & 5 & 2 \\ 2 & 7 & 2 \\ 5 & 6 & 5 \end{bmatrix}$,
11. $\begin{bmatrix} 3 & 10 & 6 \\ 5 & -1 & 7 \\ 2 & 2 & 10 \end{bmatrix}$,
12. $\begin{bmatrix} 6 & 1 & -5 \\ 4 & 6 & -4 \\ 7 & 2 & 0 \end{bmatrix}$

4. Area of objects using determinants

1. Calculate the area of the triangle with vertices at

- (a) $(-1,-2)$, $(3,8)$ and $(9,0)$.
- (b) $(1,-2)$, $(5,6)$ and $(-9,0)$.
- (c) $(-6,-1)$, $(-5,5)$ and $(-1,2)$.
- (d) $(-6,-1)$, $(-4,-3)$ and $(-1,2)$.

- (e) $(3,-3)$, $(-4,-3)$ and $(-1,2)$.
- (f) $(3,-3)$, $(5,4)$ and $(-1,2)$.
- (g) $(-5,5)$, $(5,4)$ and $(-1,2)$.
- (h) $(-5,5)$, $(5,4)$ and $(3,-3)$.

2. Calculate the area of a parallelogram whose vertices are given by $(-5,5)$, $(-4,-3)$, $(3,-2)$ and $(2,6)$.